

Given three integers n , a and b return n^{th} magical number.
Since the answer may be very large return $10^9 + 7$.

A positive integer is magical if the number is divisible by either a or b .

$$a = 2, b = 3, n = 1$$
$$\text{ans} = 2.$$

$$n = 4, a = 2, b = 3$$
$$\text{ans} = 6.$$

Solⁿ - Brute force Approach.

check numbers one by one and count magical numbers.

→ function $n^{\text{th}}\text{Magical}(n, a, b)$:

$$c = 0$$

$$\text{num} = 1$$

while true:

$$\text{if } \text{num} \% a == 0 \text{ or } \text{num} \% b == 0 :$$

$$c++$$

$$\text{if } c == n$$

return num

$$\text{num}++$$

$$\text{T.C} = O(n \min(a, b))$$

for brute force.

T.C

↓

$$O(N \min(a, b))$$

→ It fails as
constraints go up
to 10^9 .

Optimal Approach.

→ Instead of checking numbers one by one, binary search the answer.

$$\bullet \text{Count}(x) = x/A + x/B - x/\text{LCM}(a, b).$$

• we want the smallest x such that:

$$\text{count}(x) \geq N.$$

$$L = 10^8 \approx 10^9$$

$$R = 10^9$$

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⇒ Binary Search + LCM.

function nthmagical (n, a, b):

$$\text{MOD} = 1e9 + 7$$

$$l = 1$$

$$r = n * \min(a, b)$$

$$\text{lcm} = (a * b) / \text{gcd}(a, b)$$

while $l < r$:

$$\text{mid} = (l + r) / 2$$

$$c = \text{mid}/a + \text{mid}/b - \text{mid}/\text{lcm}$$

if $c < N$

$$l = \text{mid} + 1$$

else:

$$r = \text{mid}$$

$$T.C = O(\log(n * \min(a, b)))$$

return $l \% \text{MOD}$.

Dry Run:-

$$n = 1$$

$$(1) \text{gcd}(2, 3) = 1$$

$$a = 2$$

$$\text{lcm}(2, 3) = 6$$

$$b = 3$$

$$(2) l = 1$$

$$r = n * \min(a, b) = 1 * 2 = 2$$

then, (3) $1/2 = 0$

$$1/3 = 0$$

$$c = 0 + 0 - 0 = 0$$

$$1/6 = 0$$

then,

$$l = 2$$

$$\text{mid} = 2$$

$$r = 2$$

} ⇒

$$2/2 = 1$$

$$2/3 = 0$$

$$2/6 = 0$$

$$c = 1 + 0 - 0 = 1$$

count $\geq N$.

∴ Ans = 2.